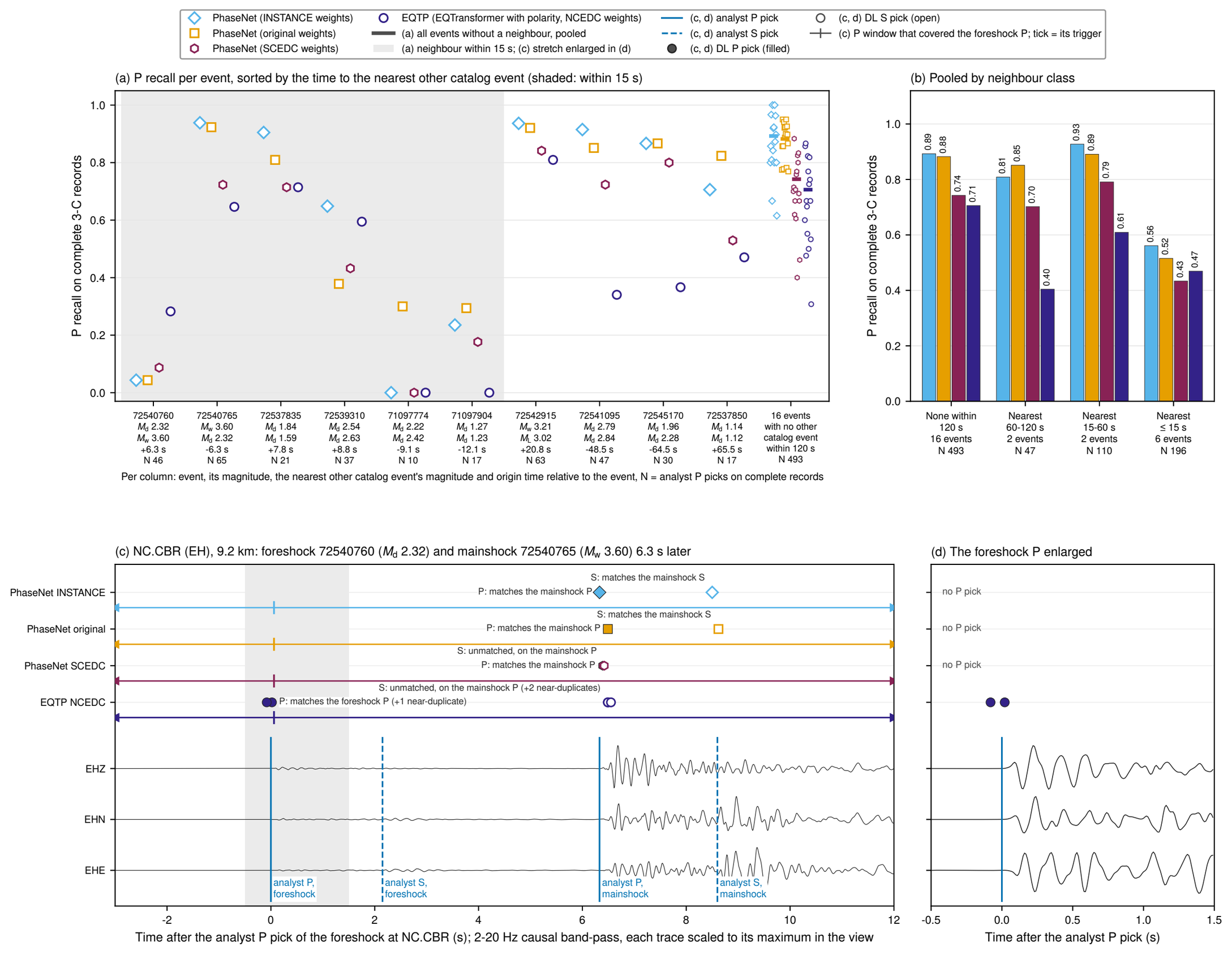
PhaseNet (INSTANCE weights)
PhaseNet (original weights)
PhaseNet (SCEDC weights)
EQTP (EQTransformer with polarity, NCEDC weights)
(a) all events without a neighbour, pooled
(a) neighbour within 15 s; (c) stretch enlarged in (d)
(c, d) analyst P pick
(c, d) analyst S pick
(c, d) DL P pick (filled)
(c, d) DL S pick (open)
(c) P window that covered the foreshock P; tick = its trigger
(a) P recall per event, sorted by the time to the nearest other catalog event (shaded: within 15 s)
P recall on complete 3-C records
72540760 Md 2.32 Mw 3.60 +6.3 s N 46
72540765 Mw 3.60 Md 2.32 -6.3 s N 65
72537835 Md 1.84 Md 1.59 +7.8 s N 21
72539310 Md 2.54 Md 2.63 +8.8 s N 37
71097774 Md 2.22 Md 2.42 -9.1 s N 10
71097904 Md 1.27 Md 1.23 -12.1 s N 17
72542915 Mw 3.21 ML 3.02 +20.8 s N 63
72541095 Md 2.79 Md 2.84 -48.5 s N 47
72545170 Md 1.96 Md 2.28 -64.5 s N 30
72537850 Md 1.14 Md 1.12 +65.5 s N 17
16 events with no other catalog event within 120 s N 493
Per column: event, its magnitude, the nearest other catalog event's magnitude and origin time relative to the event, N = analyst P picks on complete records
(b) Pooled by neighbour class
P recall on complete 3-C records
0.89
0.88
0.74
0.71
0.81
0.85
0.70
0.40
0.93
0.89
0.79
0.61
0.56
0.52
0.43
0.47
None within 120 s 16 events N 493
Nearest 60-120 s 2 events N 47
Nearest 15-60 s 2 events N 110
Nearest ≤ 15 s 6 events N 196
(c) NC.CBR (EH), 9.2 km: foreshock 72540760 (Md 2.32) and mainshock 72540765 (Mw 3.60) 6.3 s later
PhaseNet INSTANCE
PhaseNet original
PhaseNet SCEDC
EQTP NCEDC
S: matches the mainshock S
P: matches the mainshock P
S: unmatched, on the mainshock P
S: unmatched, on the mainshock P (+2 near-duplicates)
P: matches the foreshock P (+1 near-duplicate)
EHZ
EHN
EHE
analyst P, foreshock
analyst S, foreshock
analyst P, mainshock
analyst S, mainshock
Time after the analyst P pick of the foreshock at NC.CBR (s); 2-20 Hz causal band-pass, each trace scaled to its maximum in the view
(d) The foreshock P enlarged
no P pick
no P pick
no P pick
Time after the analyst P pick (s)